Insight-V: Exploring Long-Chain Visual Reasoning with Multimodal Large Language Models

Anonymous Author

Abstract

Multimodal Large Language Models (MLLMs) have demonstrated impressive capabilities across a wide range of visual understanding tasks. However, performing complex, long-chain, step-by-step visual reasoning remains a significant challenge, largely due to the scarcity of high-quality training data and the lack of effective training paradigms. Directly training on long chain-of-thought traces often leads to degraded stability and perception, causing error propagation that undermines model reliability.

*In this paper, we present **Insight-V**, a system that addresses these limitations through two key innovations. First, we introduce a scalable, automated data generation pipeline that produces long, structured visual reasoning traces via a progressive generator. The generator emits JSON-formatted steps with explicit action signals (continue or summary), enabling the collection of diverse, high-quality reasoning paths at scale. A multi-granularity assessment mechanism filters and scores these paths to ensure data quality.*

*Second, we propose a **multi-agent training framework** that disentangles reasoning from answer summarization. A reasoning agent is trained via supervised fine-tuning on the best long-chain paths and subsequently refined with iterative Direct Preference Optimization (DPO). A separate summary agent consumes the reasoning output enclosed in `<thoughts>` tags and produces the final answer, and is trained to robustly handle both correct and flawed reasoning chains.*

*We apply Insight-V to LLaVA-NeXT-8B and a stronger Qwen2.5-7B-based MLLM. On seven visual reasoning benchmarks, Insight-V achieves an average improvement of **+7.0** points over LLaVA-NeXT-8B (40.2→47.2) and **+2.9** points over the stronger 7B baseline (48.7→51.6). Importantly, general perception capabilities such as TextVQA, DocVQA, OCRBench, and AI2D also improve, demonstrating that our approach enhances reasoning without sacrificing perception. Code and models are publicly available.*

1. Introduction

Multimodal Large Language Models (MLLMs) have made remarkable strides in understanding and reasoning over visual content [**? ? ? ? ? ? ?**]. Yet, complex visual reasoning that requires long, structured, step-by-step thinking remains a persistent bottleneck. While Chain-of-Thought (CoT) prompting [**?**] has proven transformative for text-based reasoning in large language models, extending this paradigm to the multimodal domain faces two fundamental challenges: (1) the scarcity of long-chain, high-quality visual reasoning data, and (2) the lack of effective training strategies that can leverage such data without introducing instabilities.

A naive approach to visual reasoning would be to directly supervise an MLLM on long reasoning traces. However, this conflates the reasoning process with the answering process, making the model susceptible to error propagation: a single mistake in an early reasoning step can cascade and corrupt the final answer. Moreover, optimizing on lengthy chain-of-thought sequences often degrades the model's fundamental perception capabilities, as the training signal for fine-grained visual understanding becomes diluted by the extensive reasoning steps.

To overcome these challenges, we propose **Insight-V**, a comprehensive framework for enabling long-chain visual reasoning in MLLMs. Our approach rests on three pillars:

Scalable Long-Chain Data Generation. We design a progressive data generation pipeline that leverages a strong MLLM (Qwen2-VL [**?**]) to generate structured reasoning traces. Each trace consists of sequential JSON-formatted steps, where each step contains a title, content, and an explicit `next_action` signal (*continue* or *final_answer*). By sampling multiple diverse paths per query and applying a multi-granularity assessment pipeline—using a strong LLM for answer correctness filtering and an MLLM for reasoning quality scoring—we curate a large-scale dataset of approximately 200K high-quality reasoning paths.

Multi-Agent Training Framework. Instead of training a single model end-to-end on full reasoning chains, we decompose the problem into two specialized agents. The *reasoning agent* is dedicated to generating step-by-step reasoning pro-

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cesses and is fine-tuned on the curated long-chain data. The *summary agent* receives the reasoning chain (enclosed in <thoughts> tags) and generates the final answer, and is trained to critically evaluate and summarize (or override) the reasoning. This architectural disentanglement prevents error propagation and preserves the model's general visual perception capabilities.

Iterative Preference Optimization. To further refine the quality of long-chain reasoning, we apply an iterative Direct Preference Optimization (DPO) [?] procedure to the reasoning agent. Over three rounds, the model generates preference pairs from its own outputs, and DPO is applied to push the model toward higher-quality reasoning. This iterative process consistently improves reasoning quality beyond what a single round of preference optimization can achieve.

We validate Insight-V on two strong baselines: LLaVA-NeXT-LLaMA3 [?] (8B parameters) and a custom Qwen2.5-7B-based MLLM. Across seven visual reasoning benchmarks—MMMU [?], MMMU-Pro [?], MMBench [?], MME [?], ChartQA [?], MMStar [?], and MathVista [?]—Insight-V achieves consistent and substantial improvements over the baselines. Notably, Insight-V-LLaVA improves the average reasoning score by +7.0 points (40.2→47.2), while the perception-oriented benchmarks (TextVQA, DocVQA, OCRBench, AI2D) show concurrent gains, confirming that our framework enhances reasoning without sacrificing general perception.

Our contributions are summarized as follows:

- A scalable, automated pipeline for generating long, structured visual reasoning data with progressive generation and multi-granularity assessment.
- A multi-agent MLLM framework that disentangles reasoning from summarization, improving robustness to flawed intermediate chains.
- An iterative DPO procedure that progressively enhances long-chain reasoning quality using model-generated preference pairs.
- Consistent and generalizable gains across diverse visual reasoning and perception benchmarks, demonstrated on two different model backbones.

2. Related Work

Multimodal Large Language Models. The rapid development of large language models has catalyzed a new generation of MLLMs capable of jointly processing images and text [? ? ? ? ? ? ? ?]. Representative works such as LLaVA [?] and LLaVA-1.5 [?] establish the paradigm of connecting a visual encoder to a language model via a lightweight connector. LLaVA-NeXT [?] further improves upon this by increasing image resolution and training data scale. More recent systems such as InternVL [?], LLaVA-OneVision [?], Qwen2-VL [?], and Cambrian-1 [?] push the frontier with larger datasets, stronger visual encoders, and improved training recipes. Despite impressive results on standard benchmarks, these models still struggle with tasks requiring extended step-by-step visual reasoning.

Chain-of-Thought Reasoning in Vision-Language Models. Chain-of-thought (CoT) prompting [?] has proven transformative for enhancing reasoning in language models by generating intermediate reasoning steps. In the multimodal setting, several works have explored adapting CoT-style reasoning to vision-language tasks [? ?]. However, directly applying long CoT supervision to MLLMs faces challenges: the training can be unstable, reasoning steps may entangle with answer generation, and fine-grained visual perception can degrade. Our work addresses these challenges by separating reasoning from answering and employing a specialized data generation pipeline.

Preference Optimization for LLMs and MLLMs. Reinforcement Learning from Human Feedback (RLHF) [?] has become a standard approach for aligning language models with human preferences. Direct Preference Optimization (DPO) [?] simplifies this process by directly training on preference pairs without an explicit reward model, based on the Bradley-Terry model [?]. Recent work has explored AI-generated feedback as an alternative to human annotations [?], and self-play-based fine-tuning [?] for iterative model improvement. In the multimodal domain, applying preference optimization to improve reasoning quality is an emerging area. Our iterative DPO approach builds on these ideas, using model-generated preference pairs to progressively refine the reasoning agent.

Data Generation for Visual Reasoning. High-quality training data is critical for improving MLLM reasoning. Several works have explored automatic data generation and filtering for visual tasks [? ? ? ?]. Our approach extends these efforts with a structured, multi-step generation process and a multi-granularity assessment pipeline specifically designed for long-chain reasoning data.

3. Methodology

3.1. Overview

Insight-V consists of three main components: (1) a long-chain visual reasoning data generation pipeline, (2) a multi-agent training framework with a dedicated reasoning agent and summary agent, and (3) an iterative preference optimization procedure. An overview of the model design is shown in Fig. 1, and the data generation pipeline is illustrated in Fig. 2.

3.2. Long-Chain Visual Reasoning Data Generation

A central challenge in training MLLMs for complex reasoning is the lack of large-scale, high-quality long-chain

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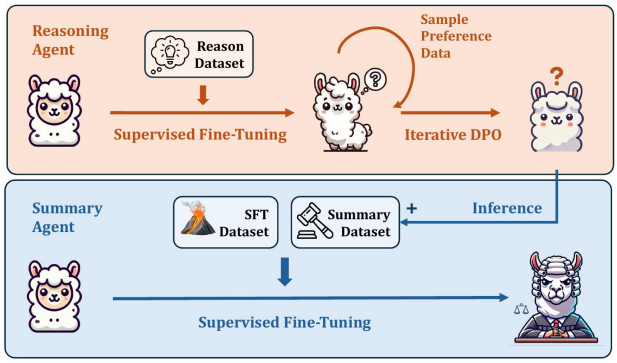

Figure 1. **Overview of Insight-V Model Design.** The reasoning agent is trained via supervised fine-tuning on long-chain reasoning data and refined with iterative DPO. The summary agent is trained on both the SFT dataset and a curated summary dataset, enabling it to generate accurate answers by leveraging or selectively overriding the provided reasoning chain.

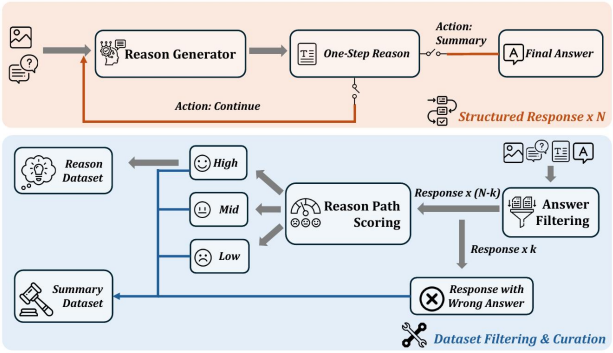

Figure 2. **Data Generation Pipeline of Insight-V.** A progressive reason generator iteratively produces structured reasoning steps. After N rounds of generation, a multi-granularity assessment pipeline filters paths by answer correctness and scores surviving paths by reasoning quality, producing a curated Reason Dataset and Summary Dataset.

reasoning data. We address this with an automated progressive data generation pipeline.

Progressive Reason Generator. Given a visual question, we prompt a strong MLLM (Qwen2-VL [?]) to generate reasoning traces in a structured JSON format. At each step, the model outputs a JSON object with three fields: `title` (a brief description of the current reasoning step), `content` (the detailed reasoning for this step), and `next_action` (either `continue` or `final_answer`). The generation process iterates until the model sets `next_action` to `final_answer` or a maximum number of steps (15) is reached, producing a complete long-chain reasoning trace. To encourage diversity and coverage of alternative reasoning paths, we sample multiple traces per query using nucleus sampling (temperature $= 0.7$, top-$p = 0.95$). The model is explicitly instructed to explore alternative answers, reconsider intermediate conclusions, and use at least three distinct reasoning approaches, producing rich and varied traces.

Multi-Granularity Assessment. The generated traces vary widely in quality. We apply a two-stage filtering and scoring pipeline to curate high-quality data:

Stage 1 – Answer Filtering: Given N generated paths per query, we use a strong LLM (Qwen2 [?]) to verify whether each path reaches the correct final answer. Paths with incorrect answers (approximately k out of N) are rejected and reserved as negative examples for the summary agent's training data. The remaining $N - k$ correct paths proceed to the second stage.

Stage 2 – Reasoning Path Scoring: Among the correct paths, we use an MLLM evaluator to score each path on reasoning quality—penalizing redundant steps, circular reasoning, and unnecessary complexity, while rewarding clear, concise, and well-structured chains. Paths are categorized into high-, mid-, and low-quality tiers. The high-quality paths are selected for the *Reason Dataset* (approximately 200K samples), while the mid-quality paths are included in the *Summary Dataset*.

Dataset Curation. The *Reason Dataset* (∼200K) contains the best reasoning chains and is used to train the reasoning agent. The *Summary Dataset* (∼1.2M) contains a mixture of: (1) optimal reasoning paths for positive training, (2) mid-quality reasoning paths, (3) reasoning paths with wrong answers for robustness training, and (4) general visual QA data to preserve perception capabilities. This diversity is crucial for training a robust summary agent.

3.3. Multi-Agent Training Framework

Reasoning Agent. The reasoning agent is a full MLLM trained to generate step-by-step reasoning processes in response to visual questions. It is first fine-tuned via Supervised Fine-Tuning (SFT) on the ∼200K samples from the Reason Dataset. The training objective is standard cross-entropy over the target reasoning tokens. The reasoning agent focuses exclusively on producing high-quality reasoning chains and does not need to produce a final answer—this separation of concerns is key to preventing the model from shortcutting the reasoning process.

Summary Agent. The summary agent is trained to receive the full reasoning chain (wrapped in `<thoughts>...</thoughts>` tags) along with the original image and question, and to produce a concise, accurate final answer. Critically, the summary agent is trained on a mixture of correct and incorrect reasoning chains, teaching it to critically evaluate whether the provided reasoning is valid and to override it when necessary. The summary agent's training data includes ∼1.2M samples: a

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mix of high-quality SFT data, paths with flawed reasoning, and general visual QA data (e.g., image captions and standard instruction-following data) to retain perception abilities. The training uses a learning rate of 2×10^{-5}.

Inference Pipeline. At inference time, the two agents operate in a pipeline. The reasoning agent first processes the image and question, generating a multi-step reasoning chain enclosed in `<thoughts>` tags. The summary agent then receives the original image, the question, and the reasoning chain, and outputs the final answer. The summary agent's prompt explicitly asks it to judge whether the reasoning is correct and to either summarize the reasoning or answer the question independently.

3.4. Iterative Preference Optimization

After SFT training, we apply iterative DPO [?] to the reasoning agent to further improve reasoning quality. DPO directly optimizes the model to prefer high-quality reasoning over lower-quality alternatives, based on the Bradley-Terry preference model [?], without requiring explicit reward model training.

Preference Pair Generation. For each training query, we generate multiple reasoning chains from the current reasoning agent using diverse sampling. We then apply the multi-granularity assessment pipeline from Sec. 3 to score these chains. For each query, we pair the highest-scoring chain (chosen) with a lower-scoring chain (rejected), creating preference pairs. Approximately 15K preference pairs are generated per DPO round.

Iterative Training. We conduct three rounds of iterative DPO. In each round, the updated reasoning agent from the previous round is used to generate new preference pairs, ensuring the training distribution remains close to the model's current behavior. This iterative approach is important: as the model improves, it generates higher-quality reasoning chains overall, and the preference pairs in later rounds capture more subtle quality differences. DPO training uses a learning rate of 5×10^{-7}. As shown in our ablation study (Sec. 4.4), iterative DPO (avg. 63.3) consistently outperforms single-pass DPO (62.7) and RLAIF [?] (62.3), demonstrating the value of the iterative approach.

Base Models. We apply Insight-V to two base models: (1) **LLaVA-NeXT-LLaMA3** [?], an 8B-parameter model with a CLIP ViT-L/14 @336px visual encoder and a 2-layer MLP connector; and (2) a **custom Qwen2.5-7B MLLM** pretrained on 558K image-caption pairs followed by instruction fine-tuning on 4M samples.

4. Experiments

4.1. Experimental Setup

Benchmarks. We evaluate on seven visual reasoning benchmarks: MMMU [?], MMMU-Pro [?], MMBench [?], MME [?], ChartQA [?], MMStar [?], and MathVista [?]. We also evaluate general perception on TextVQA [?], DocVQA [?], OCRBench [?], and AI2D [?]. Evaluation is conducted using the `lmms-eval` framework; MathVista is judged by GPT-4-0613 as per the benchmark protocol.

Baselines. We compare against a broad range of state-of-the-art 7B/8B MLLMs, including DeepSeek-VL [?], VILA-1.5 [?], Cambrian-1 [?], InternLM-XComposer2 [?], POINTS [?], IXC-2.5 [?], Bunny-LLaMA3 [?], MM-1.5 [?], MiniCPM-LLaMA3-V 2.5 [?], MiniCPM-V-2.6 [?], Qwen2-VL [?], Idefics3-LLaMA3 [?], and Ovis1.5-LLaMA3 [?].

4.2. Main Results on Visual Reasoning

Tab. 1 presents results across seven visual reasoning benchmarks. Insight-V demonstrates consistent and substantial improvements on both base models.

Insight-V-LLaVA (8B). Applied to LLaVA-NeXT-LLaMA3, Insight-V improves the average reasoning score from 40.2 to **47.2 (+7.0)**. Notable improvements include: MMMU (36.9→42.0), MMMU-Pro (13.2→21.0), MMBench (72.3→81.7), ChartQA (69.4→77.4), MMStar (43.1→57.4), and MathVista (45.9→49.8). These gains span both general reasoning (MMMU, MMBench) and domain-specific reasoning (ChartQA, MathVista), demonstrating the broad applicability of our approach. The intermediate step of adding multi-agent training alone (without DPO) already yields a significant improvement (40.2→44.5), confirming the value of the architectural disentanglement.

Insight-V (7B). On the stronger Qwen2.5-based baseline (48.7 avg.), Insight-V achieves **51.6 (+2.9)**. The improvements are consistent across all benchmarks: MMMU (47.1→50.2), MMMU-Pro (22.6→24.9), MMBench (81.3→82.3), ChartQA (75.7→81.5), MMStar (57.0→61.5), and MathVista (56.9→59.9). The smaller absolute gains on this stronger baseline reflect a higher starting point; the consistent direction of improvement confirms the generalizability of Insight-V across different model families.

4.3. General Perception Results

A key concern with long-chain reasoning training is that it may degrade a model's general visual perception capabilities. Tab. 2 shows that Insight-V not only preserves but often improves perception performance.

For Insight-V-LLaVA, general perception improves significantly: TextVQA (65.2→70.5), DocVQA (78.2→82.9), OCRBench (553→663), and AI2D (71.5→77.3). We attribute this to the summary agent's training regime, which

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Table 1. **Results on Visual Reasoning Tasks.** We conduct evaluation experiments across 7 benchmarks, covering both general reasoning and task-specific reasoning assessments. Insight-V exhibits notable effectiveness and generalizability when applied to LLaVA-NeXT and our baseline model, surpassing other state-of-the-art MLLMs by a large margin.

Model	Size	MMMU	MMMU-Pro	MMBench	MME	ChartQA	MMStar	MathVista	Average
DeepSeek-VL [?]	7B	35.4	-	73.5	-/-	59.1	37.1	36.1	-
VILA-1.5 [?]	8B	38.6	-	75.3	1634.9/-	-	39.7	-	-
Cambrian-1 [?]	8B	42.7	-	75.9	1547.1/-	73.3	-	49.0	-
InternLM-XComposer2 [?]	7B	41.1	-	77.6	2220.4	71.8	56.2	59.5	-
POINTS [?]	7B	51.4	-	78.0	2184.1	-	60.9	63.0	-
IXC-2.5 [?]	7B	42.9	-	79.4	2233.1	82.2	59.9	**63.7**	-
Bunny-LLaMA3 [?]	8B	43.4	-	77.2	1588.9/321.1	-	-	34.4	-
MM-1.5 [?]	7B	41.8	-	-	1514.9/346.4	78.6	-	47.6	-
MiniCPM-LLaMA3-V 2.5 [?]	8B	45.8	19.6	77.2	2024.6	-	51.8	54.3	-
MiniCPM-V-2.6 [?]	7B	49.8	**27.2**	78.0	2268.7	-	57.5	60.6	-
Qwen2-VL [?]	7B	**53.7**	-	81.0	-	**83.0**	60.7	61.4	-
Idefics3-LLaMA3 [?]	8B	46.6	22.9	77.5	1937.4	74.8	55.9	58.4	48.1
Ovis1.5-LLaMA3 [?]	8B	48.3	23.6	76.6	1948.5	76.4	57.3	63.0	49.4
LLaVA-NeXT-LLaMA3 [?]	8B	36.9	13.2	72.3	1611.1/346.0	69.4	43.1	45.9	40.2
+ Multi-Agent	8B	40.8	17.8	77.6	1603.7/469.3	74.6	52.6	47.4	44.5
+ Iterative DPO (**Insight-V-LLaVA**)	8B	42.0	21.0	81.7	1583.9/485.4	77.4	57.4	49.8	**47.2** (+7.0)
Our Base Model	7B	47.1	22.6	81.3	1573.7/482.5	75.7	57.0	56.9	48.7
+ Multi-Agent	7B	49.7	23.8	82.2	1662.2/**629.3**	81.2	58.6	58.7	50.7
+ Iterative DPO (**Insight-V**)	7B	50.2	24.9	**82.3**	**1685.1**/627.0	81.5	**61.5**	59.9	**51.6** (+2.9)

Table 2. **Results on other multimodal benchmarks.** Insight-V enhances reasoning capabilities without compromising general visual perception and even achieves improvements on benchmarks requiring perception ability more.

Model	TextVQA	DocVQA	OCRBench	AI2D
LLaVA-NeXT-LLaMA3	65.2	78.2	553	71.5
+ Multi-Agent	68.9	81.8	631	75.7
+ Iterative DPO (**Insight-V-LLaVA**)	**70.5**	**82.9**	**663**	**77.3**
Our Base Model	75.4	90.2	713	79.7
+ Multi-Agent	**77.0**	91.4	**738**	**80.1**
+ Iterative DPO (**Insight-V**)	76.8	**91.5**	735	79.8

mixes reasoning data with general visual QA data, reinforcing visual perception while training for reasoning. For the stronger 7B model, perception results remain strong and largely competitive, with OCRBench at 735 and TextVQA at 76.8.

4.4. Ablation Studies

Effect of Multi-Agent Design (??). We ablate the multi-agent design choice against several alternatives on our 7B base model (evaluated on MMMU, ChartQA, MathVista, and MMStar):

- *Baseline*: The unmodified base model (avg. 59.2).
- *Vanilla Direct SFT*: SFT on the full reasoning chains without agent decomposition (avg. 60.6).
- *Multi-Turn Supervised*: A multi-turn format where reasoning and answering are in separate turns but use the same model (avg. 61.0).
- *Summary Agent Only*: Only the summary agent is trained (avg. 59.8).
- *Multi-Agent (Ours)*: Separate reasoning and summary agents (avg. **62.1**).

The multi-agent design outperforms all alternatives, with the gap over direct SFT (+1.5) demonstrating that disentangling reasoning from answering is critical. The multi-turn format—which applies the same model for both tasks—falls short of the multi-agent approach, suggesting that the specialization enabled by separate training objectives is key.

Table 3. **Ablations on the Insight-V Design Choice.** The multi-agent design outperforms other configurations, highlighting the critical role of reasoning and summarization decomposition.

Model	MMMU	ChartQA	MathVista	MMStar	Avg
Baseline	47.1	75.7	56.9	57.0	59.2
Vanilla - Direct SFT	47.0	79.2	57.6	58.4	60.6
Multi-Turn Supervised	48.1	79.6	57.9	58.2	61.0
Summary Agent Only	47.5	76.3	57.3	57.9	59.8
Multi-Agent	49.7	81.2	58.7	58.6	62.1

Effect of DPO Strategy (??). We compare different preference optimization strategies applied after the multi-agent training stage:

- *Multi-Agent only* (avg. 62.1): No preference optimization.
- + *RLAIF* [?] (avg. 62.3): AI feedback-based reinforcement learning.
- + *DPO* (avg. 62.7): Single-round DPO with model-generated pairs.
- + *Iterative DPO* (avg. **63.3**): Three rounds of iterative DPO (ours).

Iterative DPO achieves the highest performance, outperforming single-pass DPO by 0.6 points and RLAIF by 1.0 points. Notably, MathVista improves from 58.7 (multi-agent only) to 59.9 (iterative DPO), and MMStar improves from 58.6 to 61.5, demonstrating that iterative preference optimization meaningfully advances complex reasoning skills.

Effect of Training Data Volume (Fig. 3). We analyze the impact of training data volume on performance. As shown

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Table 4. **Ablations on the DPO training strategy.** Iterative DPO progressively enhances the model's reasoning capabilities, leading to improved performance.

Model	MMMU	ChartQA	MathVista	MMStar	Avg
Insight-V (Multi-Agent)	49.7	81.2	58.7	58.6	62.1
+ RLAIF	49.5	81.4	59.1	59.2	62.3
+ DPO	50.8	80.8	59.3	59.9	62.7
+ Iterative DPO	50.2	81.5	59.9	61.5	63.3

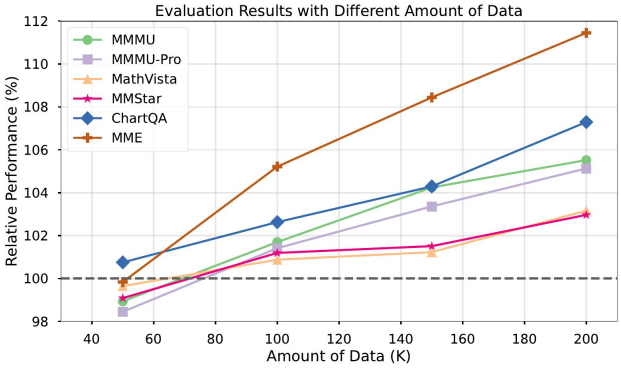

Figure 3. **Ablations on the amount of training data.** Performance (relative to 40K baseline) across five benchmarks improves consistently as training data increases from 40K to 200K samples.

in Fig. 3, model performance improves consistently as the amount of training data increases from 40K to 200K samples. The gains are most pronounced on benchmarks requiring complex reasoning (MME, ChartQA), while perception-oriented benchmarks show more modest improvements. Performance appears to plateau around 150-200K samples, suggesting that quality rather than quantity becomes the limiting factor beyond this scale.

4.5. Qualitative Analysis

Fig. 4 presents a qualitative comparison between Chain-of-Thought (CoT) inference, Insight-V (Vanilla Direct SFT), and Insight-V (Multi-Agent). Given a challenging question about marginal revenue from a table, the CoT approach arrives at an incorrect answer by misidentifying key values. The Insight-V vanilla approach (direct SFT) likewise produces reasoning that leads to an incorrect answer by making an error in option comparison. In contrast, the Insight-V multi-agent approach produces a systematic, multi-step reasoning chain: the reasoning agent carefully identifies key information, analyzes the data, and cross-checks options; the summary agent then synthesizes this into the correct final answer. This example illustrates how the disentanglement of reasoning and summarization prevents error propagation and enables more reliable visual reasoning.

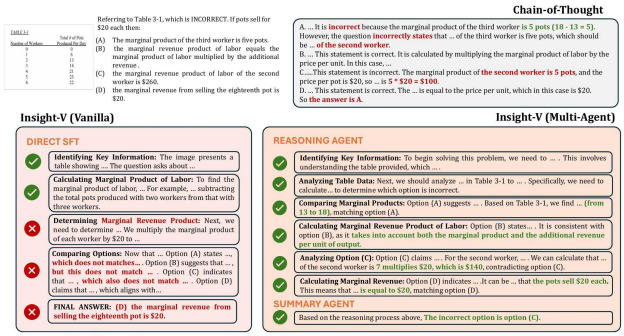

Figure 4. **Qualitative Results of Insight-V.** Comparison of Chain-of-Thought, Insight-V (Vanilla), and Insight-V (Multi-Agent) on a complex table-based reasoning question. The multi-agent approach systematically identifies key information and cross-checks options, arriving at the correct answer.

5. Conclusion

We presented Insight-V, a framework for enabling long-chain visual reasoning in MLLMs through scalable data generation and a multi-agent training paradigm. Our key finding is that *separating reasoning from answering* is critical: a dedicated reasoning agent, trained on high-quality long-chain data and refined with iterative preference optimization, substantially improves visual reasoning performance without sacrificing general perception capabilities. The summary agent, trained to robustly handle both correct and flawed reasoning chains, acts as an error-correction layer that prevents error propagation.

Insight-V achieves consistent and generalizable gains across multiple model backbones (+7.0 on LLaVA-NeXT-8B, +2.9 on Qwen2.5-7B) and across diverse benchmarks spanning general reasoning, mathematical reasoning, chart understanding, and visual perception. Our ablation studies confirm that each component—long-chain data, multi-agent design, and iterative DPO—contributes meaningfully to the final performance.

Future directions include scaling the data generation pipeline to larger and more diverse visual question datasets, exploring multi-agent architectures with more than two specialized agents, and investigating how the iterative preference optimization can be made more sample-efficient. We hope that Insight-V provides a strong foundation for future research on multi-step visual reasoning in MLLMs.

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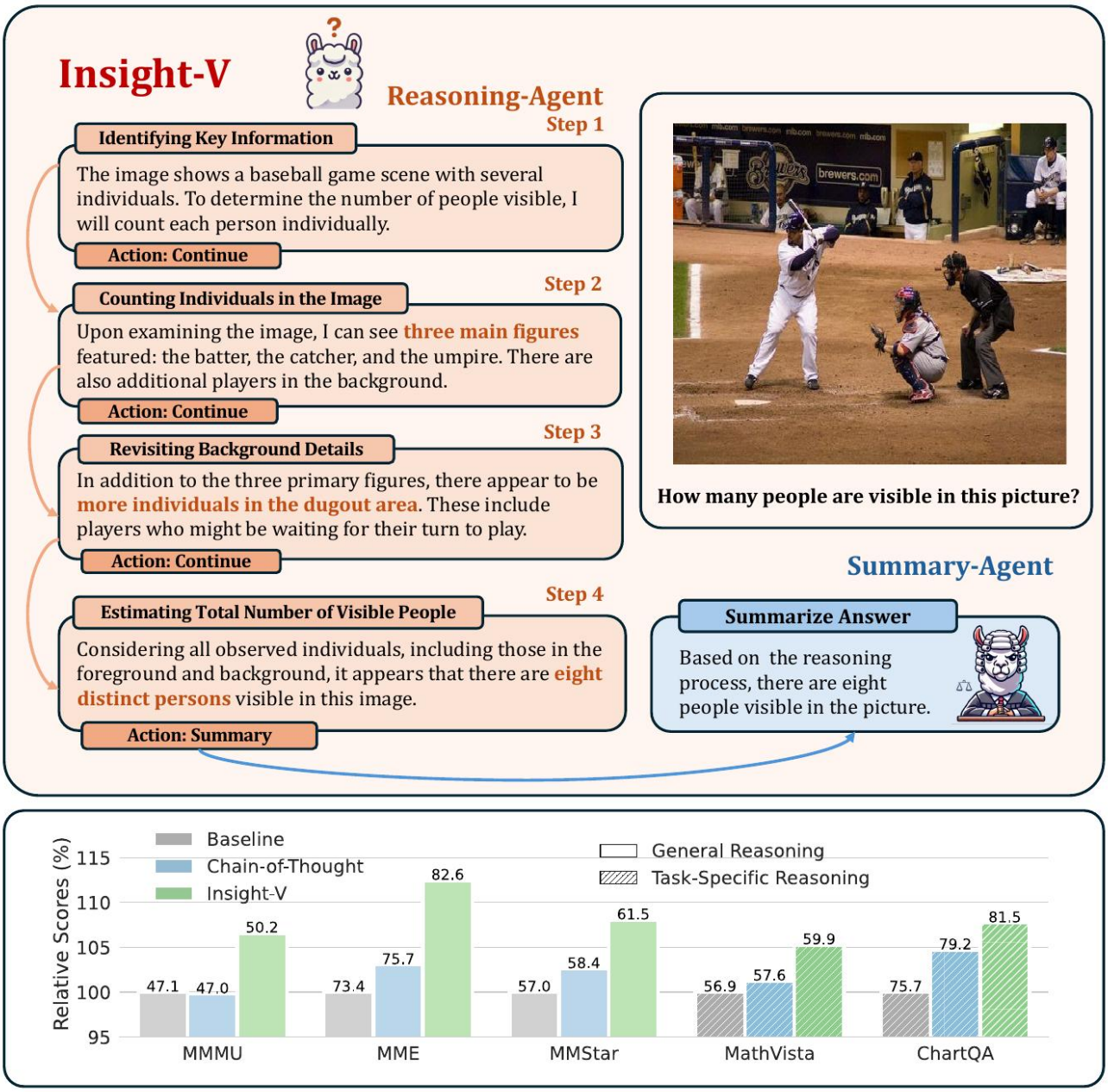

Figure 5. **Illustration and Performance of Insight-V.** (Left) Insight-V's reasoning agent decomposes visual questions into sequential reasoning steps, which are then summarized by the summary agent. (Right) Performance comparison across MMMU, MME, MMStar, MathVista, and ChartQA benchmarks, showing Insight-V's gains over the baseline and Chain-of-Thought approaches.